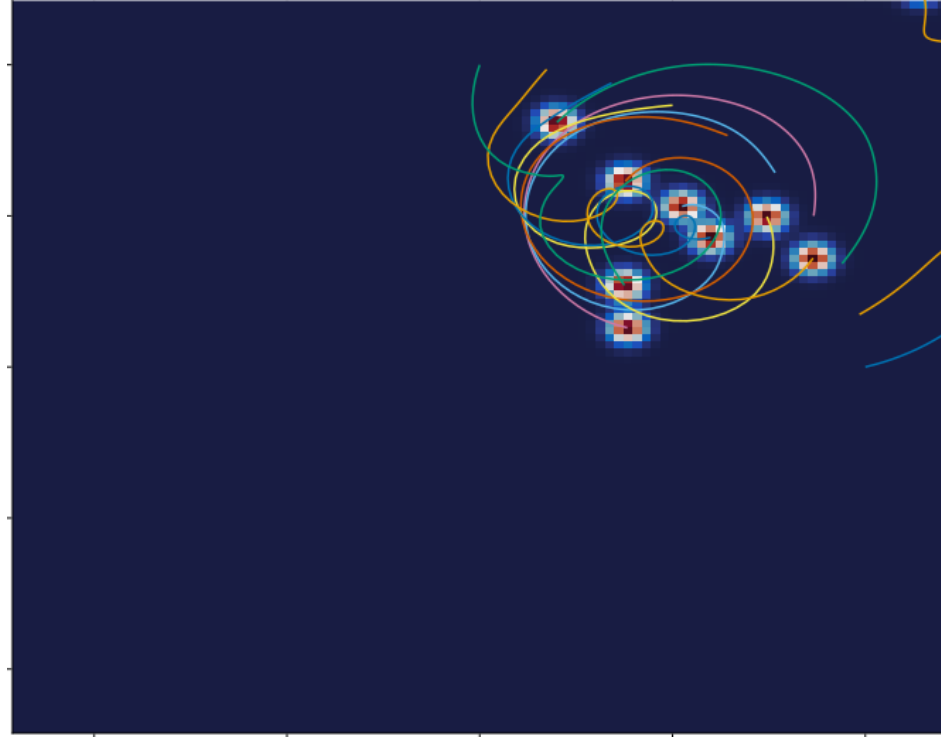


Motivation

- Vortex dynamics is the study of swirling motion in fluid flow, encompassing the formation, behavior, and interactions of vortices
- Developing nonlinear models for vortex dynamics is often complicated
- Our study aims to explore how well the Neural Networks can model for two-dimensional vortex-dominated fluid flows



Trajectory paths of ten vortices and their positions at an arbitrary time step



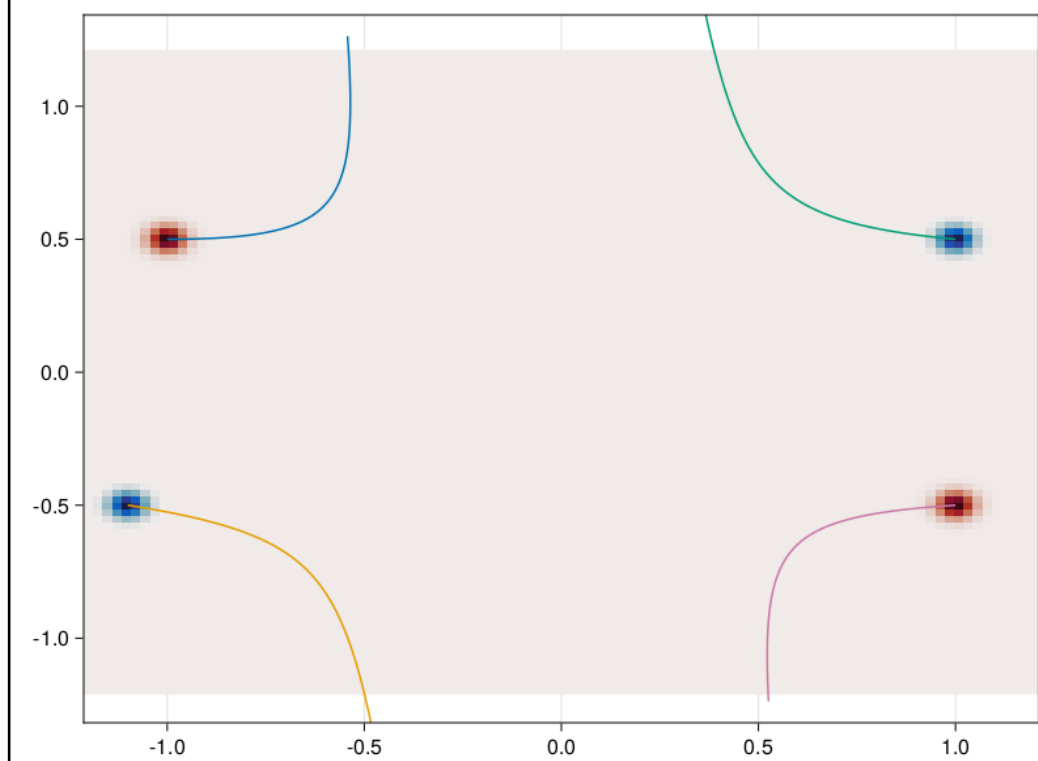
Von Kármán vortices, which can appear as long linear chains of spiral eddies that fluid flow is disturbed by an object. This image depicts the spiraling cloud pattern in an isolated Norwegian territory in the North Atlantic Ocean that occurred when winds rushing from the north encountered the Berenberg Volcano, a snow-covered peak on the eastern end of the island.

Discrete Vortex Model

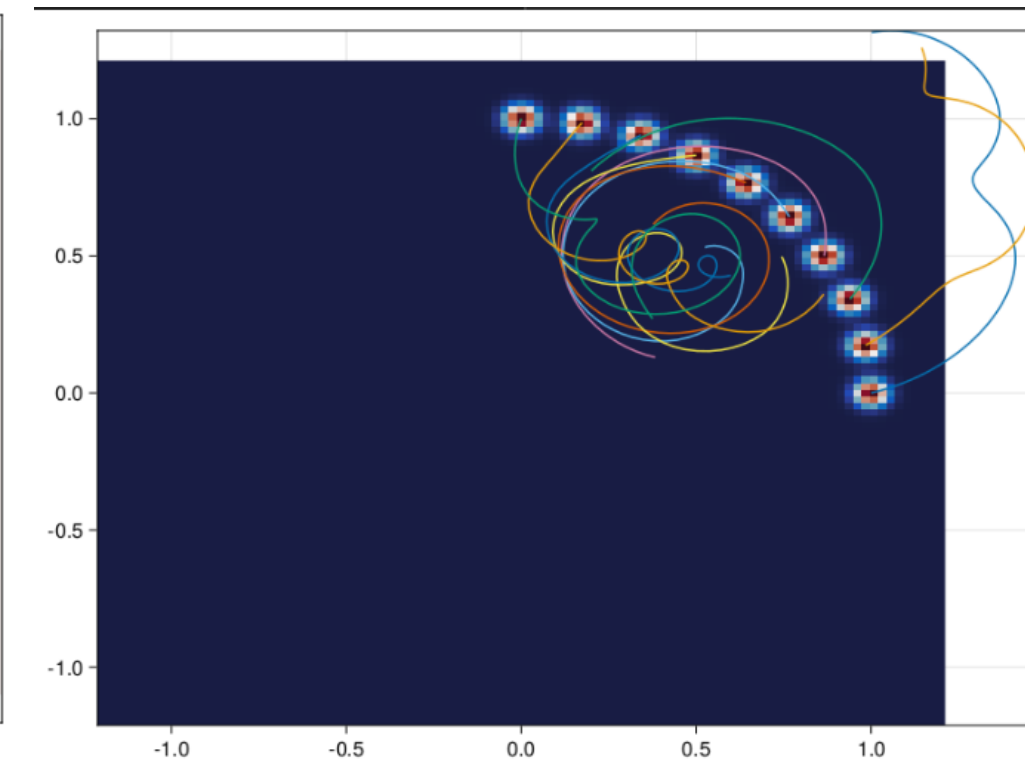
- Numerically, the fluid velocity of vortex at x_i is determined by the vorticity through the Biot-Savart Law

$$\frac{d\vec{x}_i}{dt} = - \sum_{j \neq i} \frac{(\vec{x}_i - \vec{x}_j)^\perp \Gamma_j}{|\vec{x}_i - \vec{x}_j|^2 2\pi}$$

- Examples:



A pair of vortex dipoles colliding

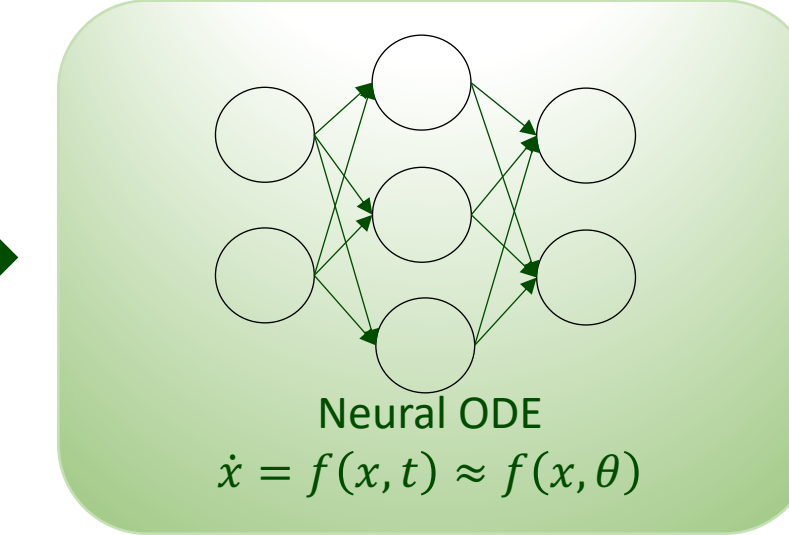


A roll-up of a line of constant-sign vortices

Neural ODEs

- Neural Ordinary Differential Equations (NODEs) can be applied to model a dynamical system with the aim of generating a trajectory that corresponds to a given initial condition.

x



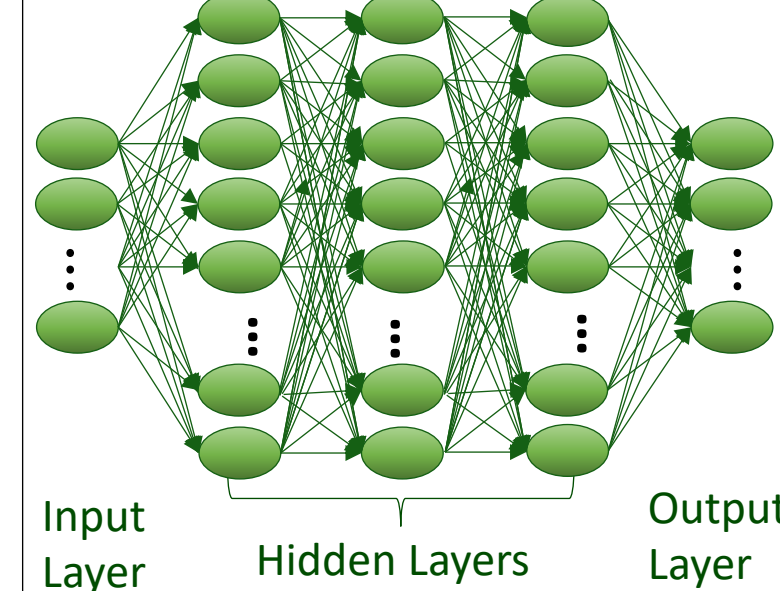
$\frac{dx}{dt}$

θ denote the all the weights and bias terms in the NN

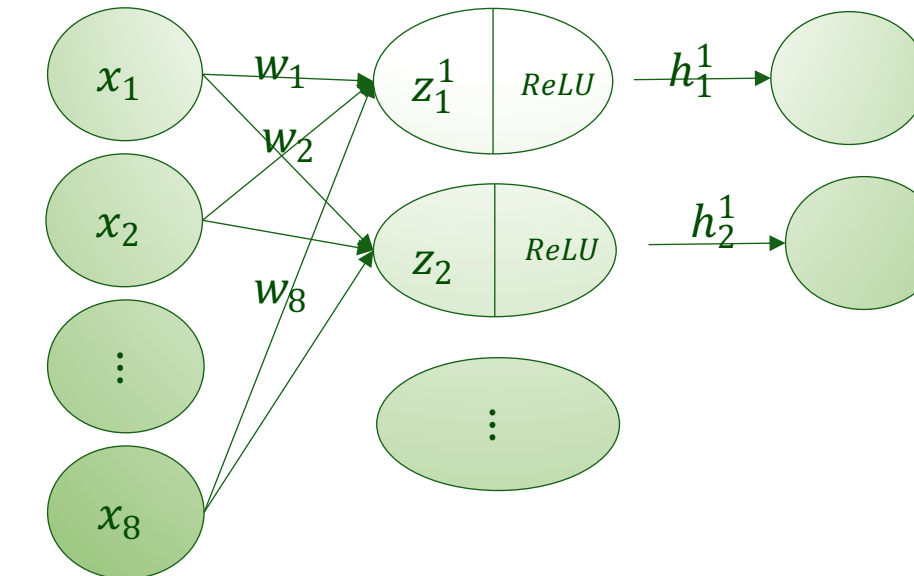
- The objective is to train the neural network to produce a reliable approximation of the trajectory derivative at each point along the trajectory.

Neural ODE Model

- Neural network



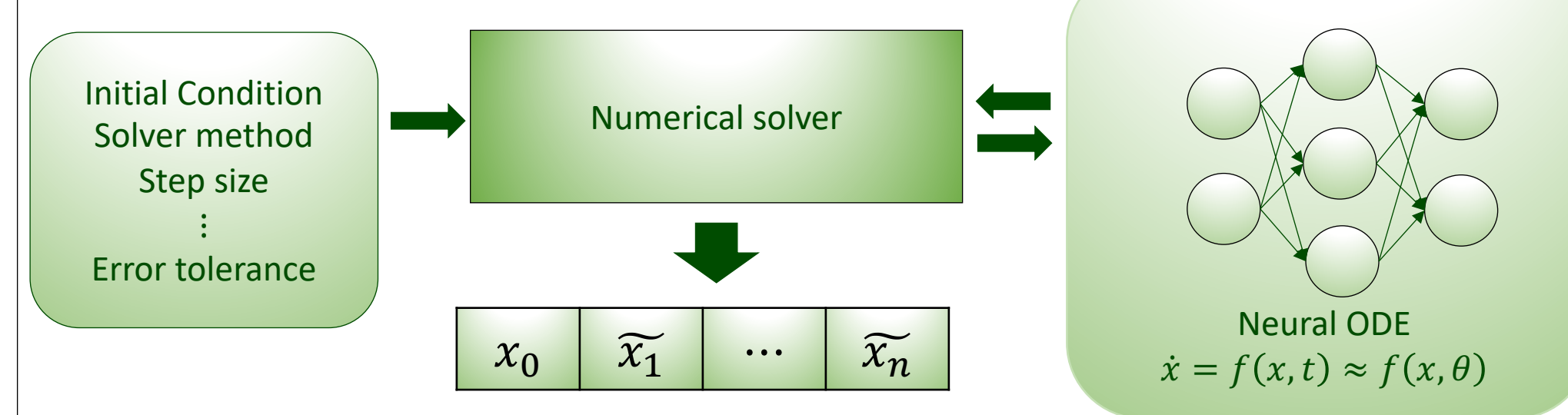
Input Layer Hidden Layers Output Layer



$$z_{(500,1)}^1 = W_{(500,8)}^T x_{(8,1)} + b_{(500,1)} \\ h_{500,1}^1 = \text{ReLU}(z_{500,1}^1)$$

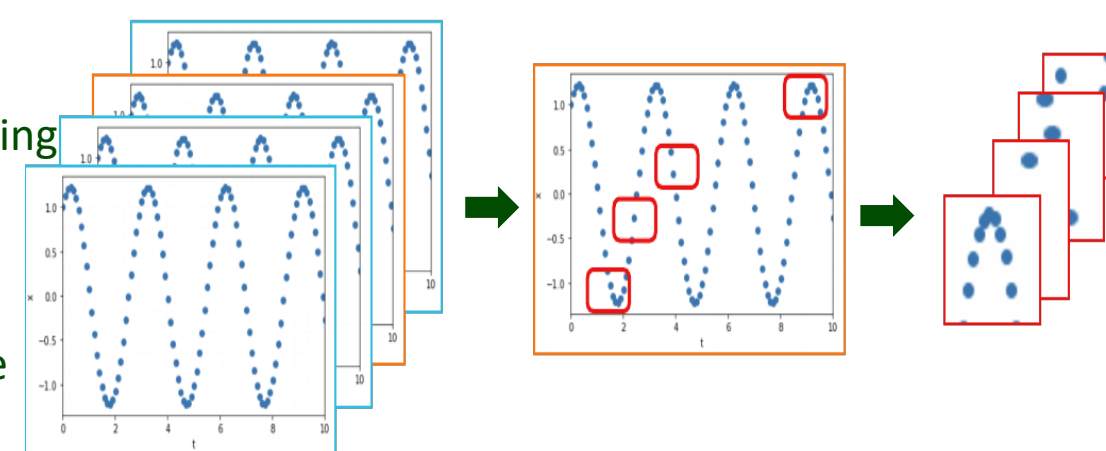
Layer	Dimension	Activation Function
Input(x)	8	
Hidden 1,2,3	800	ReLU
Output($\frac{dx}{dt}$)	8	

- NODE Model



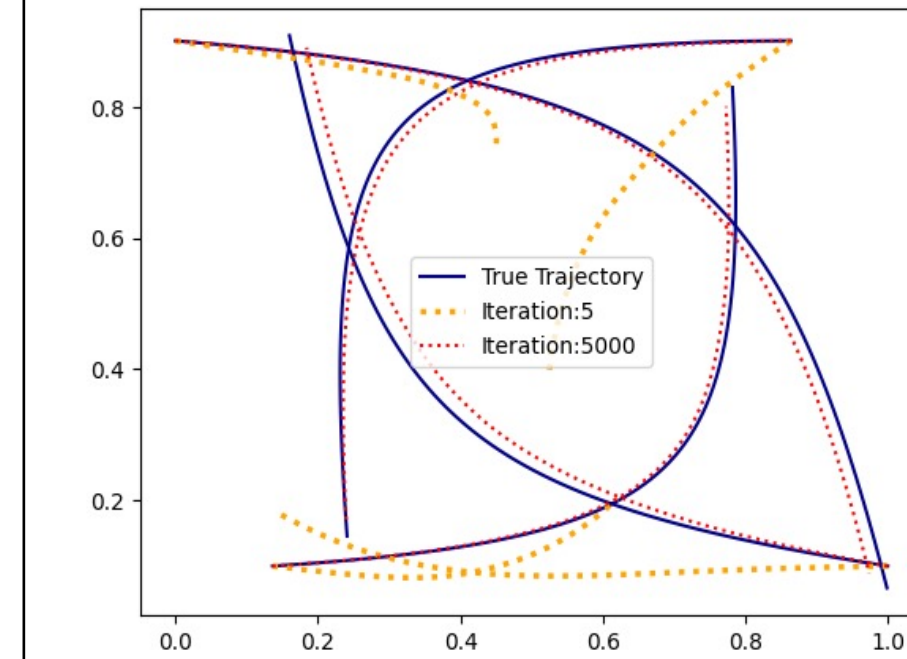
- Train

- Forward pass
Choose a batch of data (x) and simulate using NODE model ($\tilde{x}$)
- Loss computation
 $Loss = \|\|x - \tilde{x}\|_1\|$
- Backward pass by taking the gradient of the Loss w.r.t. neural network parameter θ
- Update the θ using optimization algorithm to minimize the Loss



Results

- Model convergence of pair of vortex dipoles colliding based on NODE model

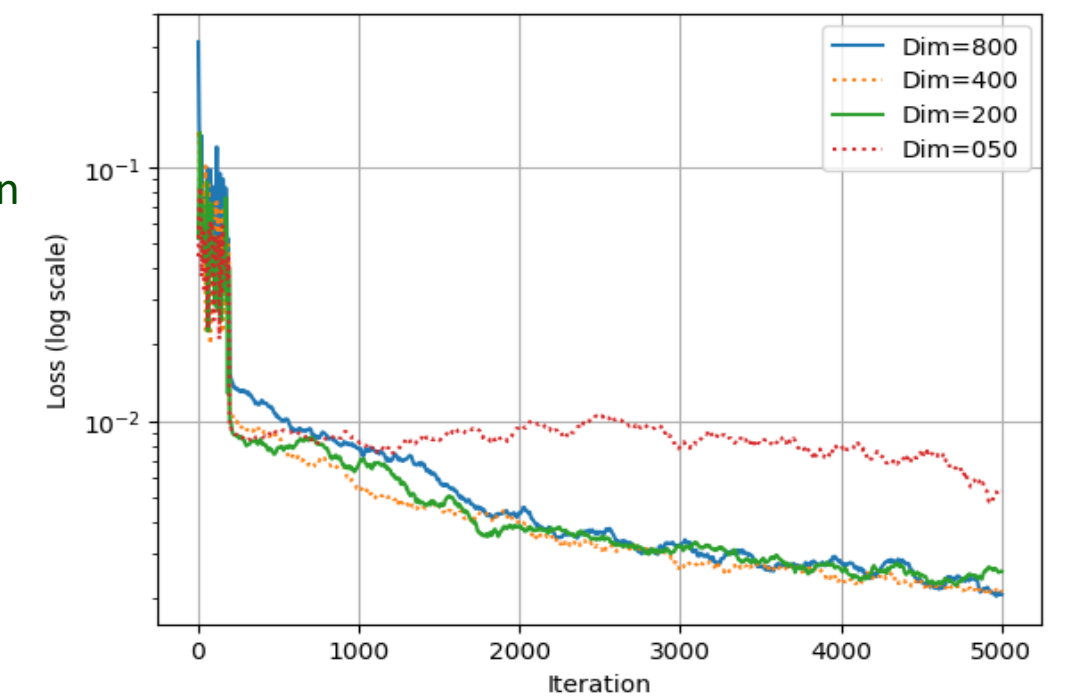


True trajectory vs model prediction at iteration 5 and iteration 5000

System Info:

- No of vortices: 4
- Γ values: 1.0, -1.0
- Iterations: 5000
- No of time steps: 100
- Solver: Dopri5

Moving average of the loss value vs iteration for different size of network

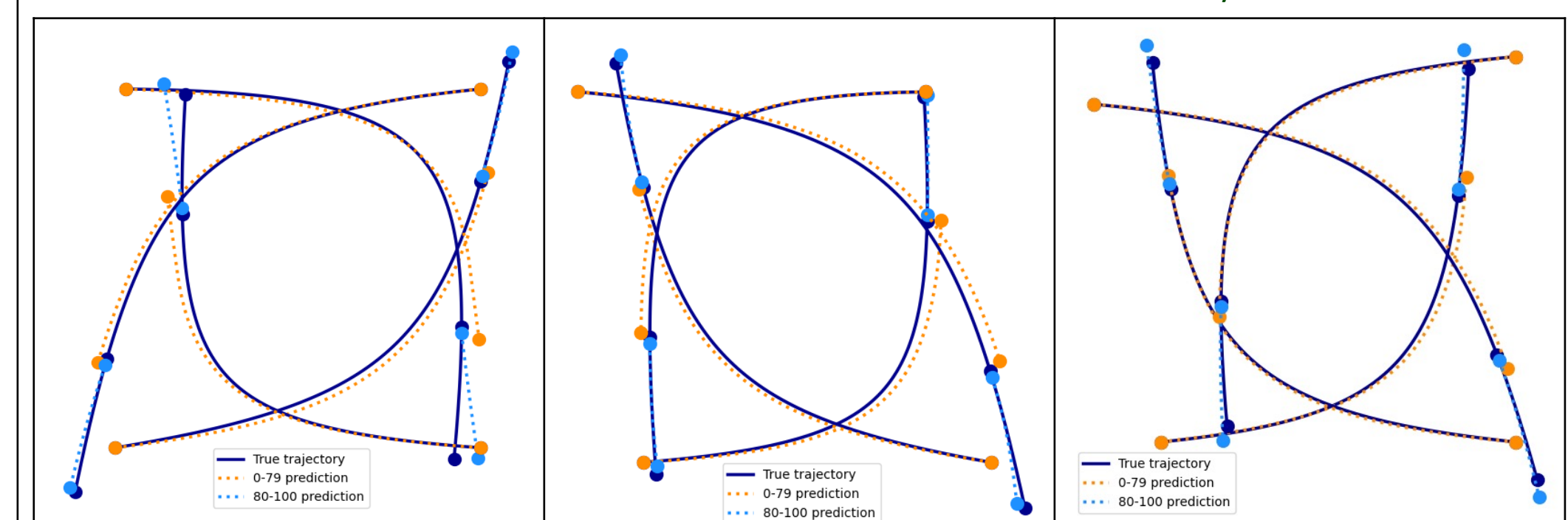


- Model prediction for future time steps

Training Validation details

- Training time steps: 0 -> 80
- Validating time steps: 0 -> 100
- Testing time steps: 80 -> 100

Hidden Layer Dimension : 50 Hidden Layer Dimension : 200 Hidden Layer Dimension : 800



Conclusion and Future work

- NODEs can closely approximate the vortex dynamics based on vortex trajectories
- One of the limitation is the method is computationally expensive so we can apply a dimensionality reduction method to reduce the input dimension of the data.

References:

- Chen, Ricky T. Q., et al. "Neural Ordinary Differential Equations." *ArXiv:1806.07366 [Cs, Stat]*, 13 Dec. 2019, arxiv.org/abs/1806.07366.
- Nitsche, M. "Vortex Dynamics." *Encyclopedia of Mathematical Physics*, 2006, pp. 390–399, <https://doi.org/10.1016/b0-12-512666-2/00254-6>.
- Legaard, Christian Møldrup, et al. "Constructing Neural Network-Based Models for Simulating Dynamical Systems." *ACM Computing Surveys*, 16 Nov. 2022, <https://doi.org/10.1145/3567591>.
- Linot, Alec J., and Michael D. Graham. "Data-Driven Reduced-Order Modeling of Spatiotemporal Chaos with Neural Ordinary Differential Equations." *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 32, no. 7, 1 July 2022, p. 073110, arxiv.org/abs/2109.00060, <https://doi.org/10.1063/5.0069536>.
- Linot, Alec J., et al. "Stabilized Neural Ordinary Differential Equations for Long-Time Forecasting of Dynamical Systems." *Journal of Computational Physics*, vol. 474, 1 Feb. 2023, p. 111838-east-1&originCreation=20230106185658, <https://doi.org/10.1016/j.jcp.2022.111838>.
- <https://earthobservatory.nasa.gov/images/77654/von-karman-vortices-in-the-greenland-sea>